Efficiency and resource measurement

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

Before you start

Why should two models be evaluated on the same examples and subgroup mix before comparing their accuracy and speed?

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Calculate throughput and amortised time per example from repeated timings. Try the recall before opening notes.

Your learning target

I can calculate **throughput** and amortised time per example from repeated timings.

I can count dense-network parameters and distinguish storage from **peak memory**.

I can compare resource use under fixed quality and measurement conditions.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Dataset shift in neural models. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Transfer learning. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Sequence representations. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Calculate **throughput** and amortised time per example from repeated timings

Count dense-network parameters and distinguish storage from **peak memory**

Compare resource use under fixed quality and measurement conditions

Retrieve what you already know

> Why should two models be evaluated on the same examples and subgroup mix before comparing their accuracy and speed?

My starting explanation

This week's end goal

By the end, I can explain efficiency and resource measurement using a traced example and a stated limitation.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Efficiency and resource measurement case. Separate observations, given facts and assumptions before choosing an explanation.

A neural model can be faster yet fail the task's accuracy requirement. **Training** cost and **prediction** cost are different: **training** updates parameters, while inference uses the fitted model.

Compare quality and resources together under a declared test, rather than calling a model efficient from one small number.

Use these invented classroom timings for models A and B. Each timed pass **process**es the same 100 pre**process**ed examples sequentially, one at a time, on the same device and precision.

Start before the first **prediction** and stop after the final **prediction** completes. Loading and pre**process**ing are excluded for both. One untimed warm-up pass precedes three timed passes; the warm-up work must still be reported.

Measurement	Model A	Model B
Pass 1 seconds	2.0	1.2
Pass 2 seconds	2.2	1.1
Pass 3 seconds	1.8	1.3
Correct on 100 validation cases	92	90

The validation cases are the same for both models and are scored once each; repeating timing passes does not create 300 independent quality examples. No actual benchmark was run to produce this table.

List the given facts and one fact that is not supplied.

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Efficiency and resource measurement. Connect each labelled input, deciding step and result.

Sort each set of three times and take the middle value. A’s median is 2.0 seconds per 100 examples; B’s is 1.2 seconds. Dividing by 100 gives amortised times of 0.020 and 0.012 seconds per example, or 20 and 12 milliseconds.

Median-pass **throughput** is 100 / 2.0 = 50 examples per second for A and about 83.33 for B.

```
from math import isfinite
from statistics import median

def timing_summary(seconds, count):
    if count <= 0 or not seconds:
        raise ValueError("Need examples and timed passes")
    if any(not isfinite(t) or t <= 0 for t in seconds):
        raise ValueError("Times must be finite and positive")
    middle = median(seconds)
    ms_per_example = 1000 * middle / count
    examples_per_second = count / middle
    return middle, ms_per_example, examples_per_second

print(timing_summary([2.0, 2.2, 1.8], 100))
print(timing_summary([1.2, 1.1, 1.3], 100))
```

The code assumes a positive integer example count and numeric times. A **output**s (2.0, 20.0, 50.0); B **output**s approximately (1.2, 12.0, 83.33). Retain full precision until display.

The time reduction is (20 - 12) / 20 = 40 percent, while **throughput** rises by about 66.67 percent. These are different denominators.

Amortised time does not reveal the slowest request or a **latency** percentile. In a batched or concurrent workload, total time divided by count is not an individual request's waiting time.

Record per-request timings if the requirement concerns **latency**; keep batch size and concurrency explicit.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Efficiency and resource measurement. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

With at least 91 correct out of 100 and at most 25 ms amortised time required, which model qualifies? Calculate B's time reduction relative to A and explain why the percentage **throughput** increase differs.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Change one thing in the pictured case

Change one thing

Change: Lower the accuracy requirement to90% before a new selection exercise.

Predicted result

Prediction: Both become eligible; Q has the smaller measured mean amortised time.

Why the result follows

Reason: The acceptance contract changed; Q's measured accuracy did not improve.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Efficiency and resource measurement to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Latency	Elapsed time for one request from a declared start to completed response
Throughput	The number of completed examples processed per unit time
Parameter storage	The bytes used to store model weights and biases at a declared precision
Peak memory	The largest amount of memory in use during a measured operation

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

Follow the worked trace for Efficiency and resource measurement, predict a step, then compare with its explanation.

A faster result can miss the requirement

MODEL	QUALITY REQUIREMENT	TIME REQUIREMENT
A: 92 of 100	At least 91 correct: pass	At most 25 ms average: pass
B: 90 of 100	At least 91 correct: fail	At most 25 ms average: pass

For this declared validation rule, only A meets both requirements. The time requirement uses amortised time from the median pass, not a per-request latency guarantee.

Worked explanation · Efficiency and resource measurement

The requirements were set before inspecting these results. B is faster under this measurement but is not eligible under the quality rule.

One hundred validation cases do not establish future accuracy; reserve independent final evaluation after choosing a model, and inspect relevant subgroup results.

If the quality requirement had instead been set in advance to at least 90 correct, with the same time limit, which models would qualify? Which would you choose if lower measured amortised time is the declared tie-breaker?

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Efficiency and resource measurement

Compare speed and quality under the same case count and acceptance contract.

The facts we will use

Supplied facts

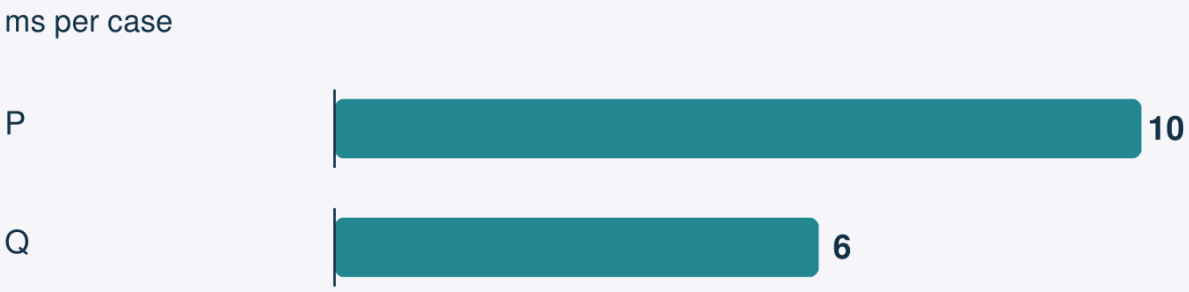
Both evaluate40 cases. P times [0.32, 0.40, 0.48]seconds, 38 correct; Q times [0.20, 0.24, 0.28]seconds, 36 correct. Requireaccuracy>=93% and mean amortised time<=12ms/case.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/31

Mean amortised time, not single-request latency



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/31

Efficiency and resource measurement

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Summarise P

Mean time0.40s; 10ms/case; accuracy 38/40=95%: passesboth.

2. Summarise Q

Mean time0.24s; 6ms/case; accuracy 36/40=90%: failsquality.

3. Choose eligible candidate

P meets the declared contract; Q is faster but ineligible.

CHANGE ONE THING

Lower the accuracy requirement to90% before a new selection exercise.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Summarise P	Mean time0.40s; 10ms/case; accuracy 38/40=95%: passesboth.
2. Summarise Q	Mean time0.24s; 6ms/case; accuracy 36/40=90%: failsquality.
3. Choose eligible candidate	P meets the declared contract; Q is faster but ineligible.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Compare models on the same workload and apply every required condition. P meets both quality and time limits; Q's speed does not compensate for missing the quality gate.

Dividing total run duration by forty gives an amortised time per case. It does not measure the wait for one isolated request, which needs its own timed experiment.

What this example does not establish

Keep the stated limit

Record hardware, batching, warm-up and variability; these supplied timings are toy

measurements, not actual benchmark runs.

Keep the stated limit

Read the labelled picture: Why is 10ms/case not necessarily a single-request **latency**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: Lower the accuracy requirement to90% before a new selection exercise. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Efficiency and resource measurement claim and explain a repair using the supplied evidence.

Check the claim against the facts

A fully connected layer with m **input**s and n **output**s has m times n weights and n biases, assuming every **output** has one bias. A network with dimensions 4 to 3 to 2 has (4 times 3 + 3) + (3 times 2 + 2) = 23 parameters.

At four bytes per parameter, its raw parameter payload is 92 bytes.

Check the claim against the facts

That is not the saved file size or peak runtime memory. A file can contain metadata, and inference also needs activations and buffers. **Training** may additionally store gradients and optimiser state.

Record whether a memory measurement refers to the model, the **process**, an accelerator or the whole device. Parameter count alone does not establish speed.

Check the claim against the facts

Runtime alone does not establish energy use or carbon emissions either. Those require additional measurements and declared assumptions, such as power over time and the electricity source.

Report what you measured rather than converting fewer parameters into an unsupported energy claim.

Locate the earliest unsupported step

Count weights, biases and total parameters in the 4 to 3 to 2 network. Calculate raw parameter bytes at four bytes each. Explain why this cannot be reported as peak **training** memory or an energy measurement.

State one repair and the checks you would rerun.

Practise together

> Practise Efficiency and resource measurement with support, then explain the deciding step in your own words.

For A and B, calculate median pass time, amortised milliseconds per example and median-pass **throughput**. State the measured boundary and one **latency** claim these numbers cannot support.

> **If the quality requirement had instead been set in advance to at least 90 correct, with the same time limit, which models would qualify? Which would you choose if lower measured amortised time is the declared tie-breaker?**

> **Explain your prediction to a partner before comparing results.**

Practise with less help: Compare under requirements

Use workbook W2 for the connected example “Efficiency and resource measurement”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Efficiency and resource measurement. Proposed explanation: “Fewer parameters always means faster”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Why is 10ms/case not necessarily a single-request **latency**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Efficiency and resource measurement. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

With at least 91 correct out of 100 and at most 25 ms amortised time required, which model qualifies? Calculate B’s time reduction relative to A and explain why the percentage **throughput** increase differs.

```
from math import isfinite
from statistics import median

def timing_summary(seconds, count):
    if count <= 0 or not seconds:
        raise ValueError("Need examples and timed passes")
    if any(not isfinite(t) or t <= 0 for t in seconds):
        raise ValueError("Times must be finite and positive")
    middle = median(seconds)
    ms_per_example = 1000 * middle / count
    examples_per_second = count / middle
    return middle, ms_per_example, examples_per_second

print(timing_summary([2.0, 2.2, 1.8], 100))
print(timing_summary([1.2, 1.1, 1.3], 100))
```

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Efficiency and resource measurement. Explain why each step is justified.

Explain → apply → challenge

For A and B, calculate median pass time, amortised milliseconds per example and median-pass **throughput**. State the measured boundary and one **latency** claim these numbers cannot support.

Count weights, biases and total parameters in the 4 to 3 to 2 network. Calculate raw parameter bytes at four bytes each. Explain why this cannot be reported as peak **training** memory or an energy measurement.

A runs on one device with batch size 1; B runs on another with batch size 32. Only B's fastest trial and the winning **training** run are reported. Explain how to compare and report the results fairly.

> **Explain which evidence supports your conclusion and which claim remains unproved.**

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Image-pattern Network Investigation. Keep an annotated decision and the evidence supporting it.

Add a labelled efficiency and resource measurement **evidence** sheet to Image-pattern Network Investigation. Show how you can calculate **throughput** and amortised time per example from repeated timings.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Record model and code versions, **input** size, device, precision, batch size, concurrency, warm-up, timed boundary, repetitions and raw results.

Alternate model order where practical, check background load and wait for computation to finish before stopping a timer; queued accelerator work may finish after the host call returns.

Report cold-start behaviour separately when it matters.

A report calls B “best and greener” because it is faster, compares B’s batched device run with A’s unbatched run on a different device, and omits B’s quality result. Name the unsupported claims and design a fair comparison.

Keep **training** and inference costs separate, and include unsuccessful trials in the total experiment budget. Changes chosen using validation need untouched final evaluation.

Next week assembles a network experiment record so another person can understand and repeat the comparison.

A runs on one device with batch size 1; B runs on another with batch size 32. Only B’s fastest trial and the winning **training** run are reported. Explain how to compare and report the results fairly.

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Efficiency and resource measurement case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

A faster result can miss the requirement

MODEL	QUALITY REQUIREMENT	TIME REQUIREMENT
A: 92 of 100	At least 91 correct: pass	At most 25 ms average: pass
B: 90 of 100	At least 91 correct: fail	At most 25 ms average: pass

For this declared validation rule, only A meets both requirements. The time requirement uses amortised time from the median pass, not a per-request latency guarantee.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Evaluate a new case independently

For 50 shared validation cases, P has times 0.6, 0.5, 0.4 seconds and 47 correct; Q has 0.3, 0.4, 0.5 seconds and 45 correct. Require at least 92 percent accuracy and at most 12 ms amortised time.

Give median times, amortised times, **throughput**, accuracy and eligible models. A separate dense network has dimensions 3 to 2 to 1 with one bias per **output**: find its parameter count and raw bytes at four bytes each.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

For 50 shared validation cases, P has times 0.6, 0.5, 0.4 seconds and 47 correct; Q has 0.3, 0.4, 0.5 seconds and 45 correct. Require at least 92 percent accuracy and at most 12 ms amortised time.

Give median times, amortised times, **throughput**, accuracy and eligible models. A separate dense network has dimensions 3 to 2 to 1 with one bias per **output**: find its parameter count and raw bytes at four bytes each.

> **My independent procedure, result and limitation**

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Efficiency and resource measurement, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Compare speed and quality under the same case count and acceptance contract.

By the end, I can	How I will show it
Calculate throughput and amortised time per example from repeated timings	Explain the deciding step, show a trace or test, and state one limit.
Count dense-network parameters and distinguish storage from peak memory	Explain the deciding step, show a trace or test, and state one limit.
Compare resource use under fixed quality and measurement conditions	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Efficiency and resource measurement without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Calculate **throughput** and amortised time per example from repeated timings

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Compare speed and quality under the same case count and acceptance contract.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? P meets the declared contract; Q is faster but ineligible.

1. Summarise P

2. Summarise Q

3. Choose eligible candidate

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Choose eligible candidate P meets the declared contract; Q is faster but ineligible. Compare models on the same workload and apply every required condition.

P meets both quality and time limits; Q's speed does not compensate for missing the quality gate. Dividing total run duration by forty gives an amortised time per case.

It does not measure the wait for one isolated request, which needs its own timed experiment.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Lower the accuracy requirement to90% before a new selection exercise. Which conclusion is supported by the supplied facts?

1. Both become eligible; Q has the smaller measured mean amortised time.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Both become eligible; Q has the smaller measured mean amortised time. The acceptance contract changed; Q's measured accuracy did not improve.

Repair a missing building block

Review Dataset shift in neural models (ai-5/30). Goal: Explain how a changed **input** mix can change overall accuracy. Short practice: Did the supplied model's dim-light accuracy deteriorate?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Transfer learning (ai-5/28). Goal: Trace a reused feature extractor and a newly fitted **output** head. Short practice: Which part was trained for the new task?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Sequence representations (ai-5/25). Goal: Encode an ordered sequence with an explicit vocabulary. Short practice: Which record lets the program distinguish the first and last zeros?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

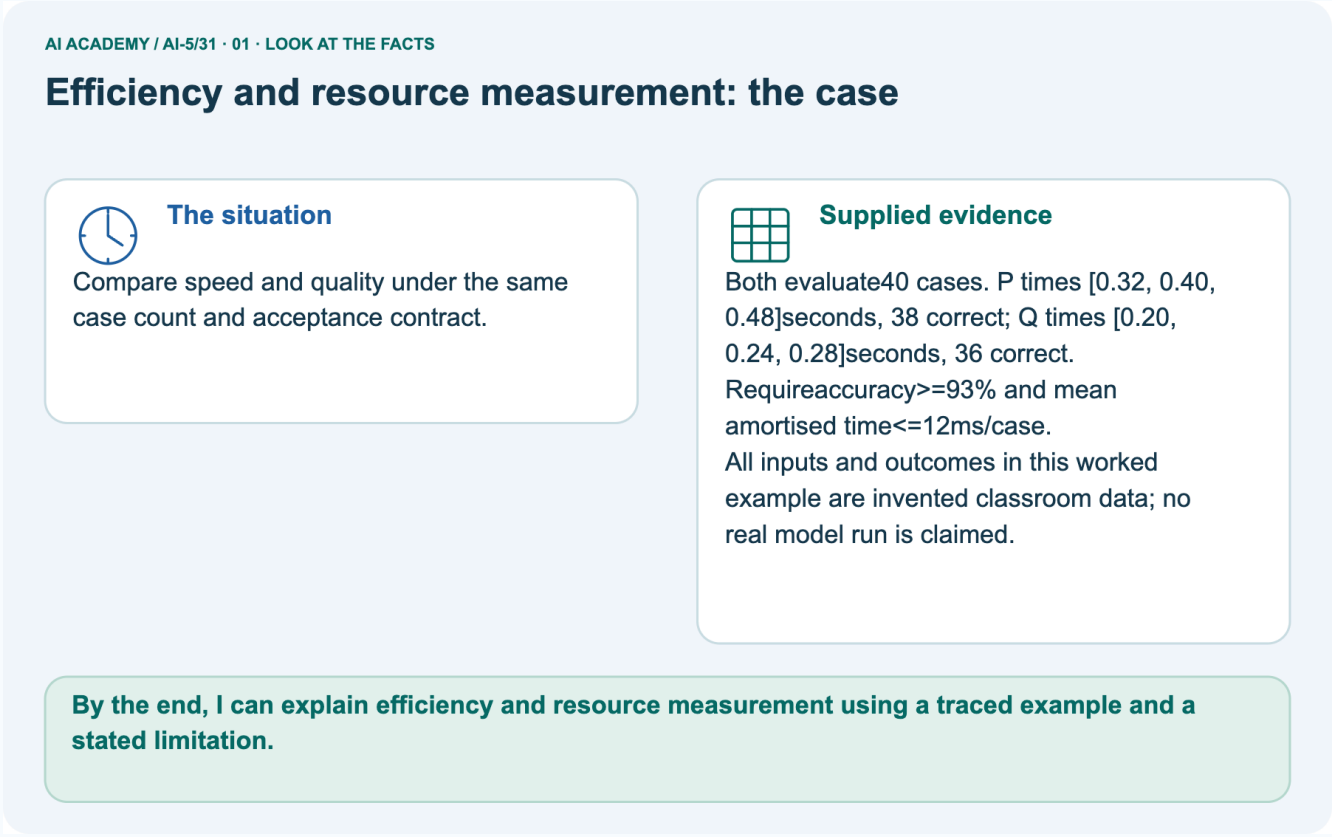
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Efficiency and resource measurement graphic using its labels, deciding evidence and relevant limitation.

01. Efficiency and resource measurement: the case



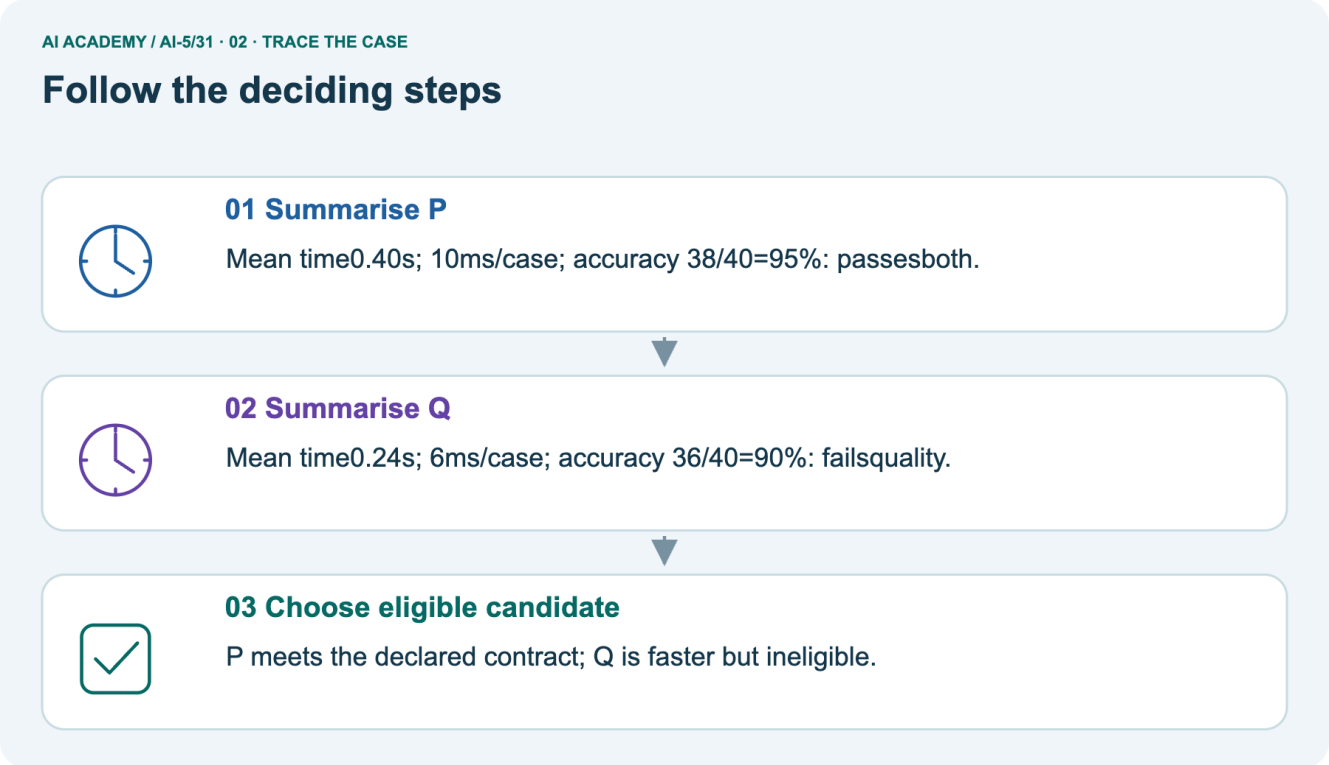
Follow the labelled picture with the matching explanation. Open the vector diagram

Compare speed and quality under the same case count and acceptance contract. Supplied facts: Both evaluate40 cases. P times [0.32, 0.40, 0.48]seconds, 38 correct; Q times [0.20, 0.24, 0.28]seconds, 36 correct.

Requireaccuracy>=93% and mean amortised time<=12ms/case. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

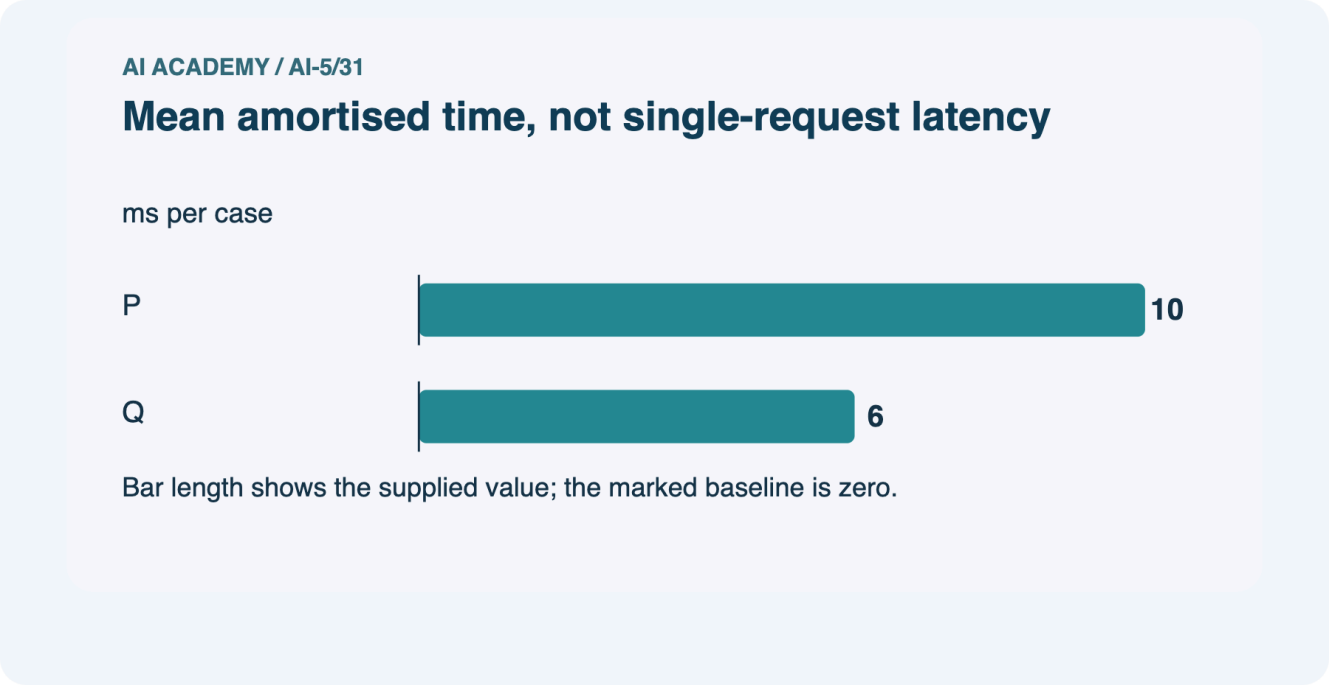


Follow the labelled picture with the matching explanation. Open the vector diagram

Summarise P: Mean time0.40s; 10ms/case; accuracy 38/40=95%: passesboth. → Summarise Q: Mean time0.24s; 6ms/case; accuracy 36/40=90%: failsquality. → Choose eligible candidate: P meets the declared contract; Q is faster but ineligible.

Point to each numbered stage and explain which recorded **input** it uses.

03. Mean amortised time, not single-request latency

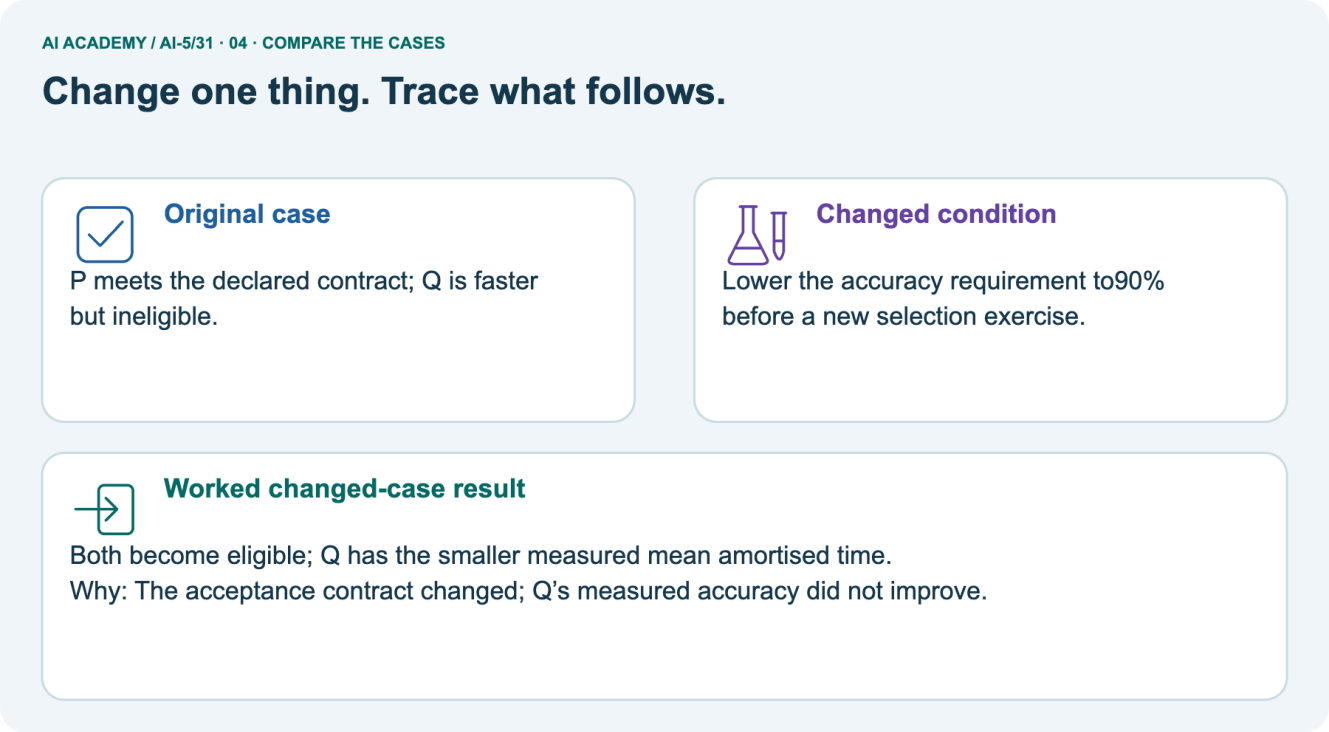


Follow the labelled picture with the matching explanation. Open the vector diagram

Mean amortised time, not single-request **latency**. Units: ms per case. P: 10. Q: 6.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



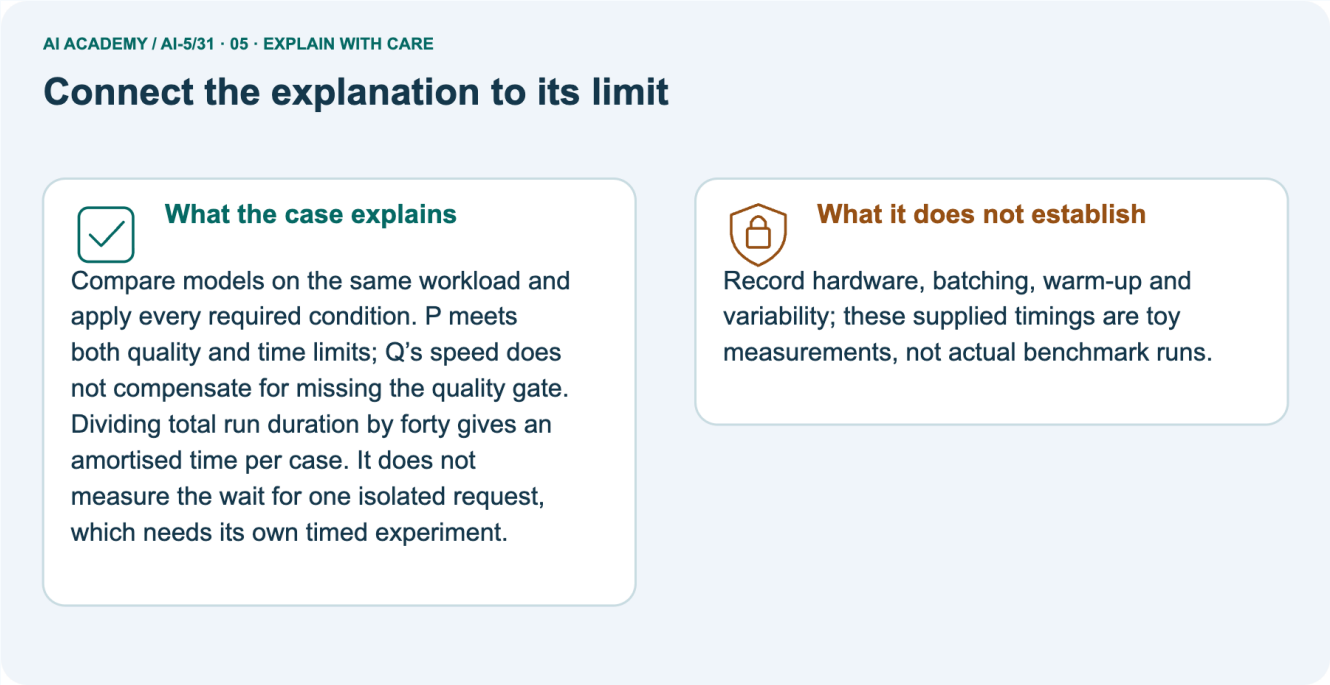
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: P meets the declared contract; Q is faster but ineligible. Changed condition: Lower the accuracy requirement to90% before a new selection exercise.

Expected result: Both become eligible; Q has the smaller measured mean amortised time. Reason: The acceptance contract changed; Q's measured accuracy did not improve.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

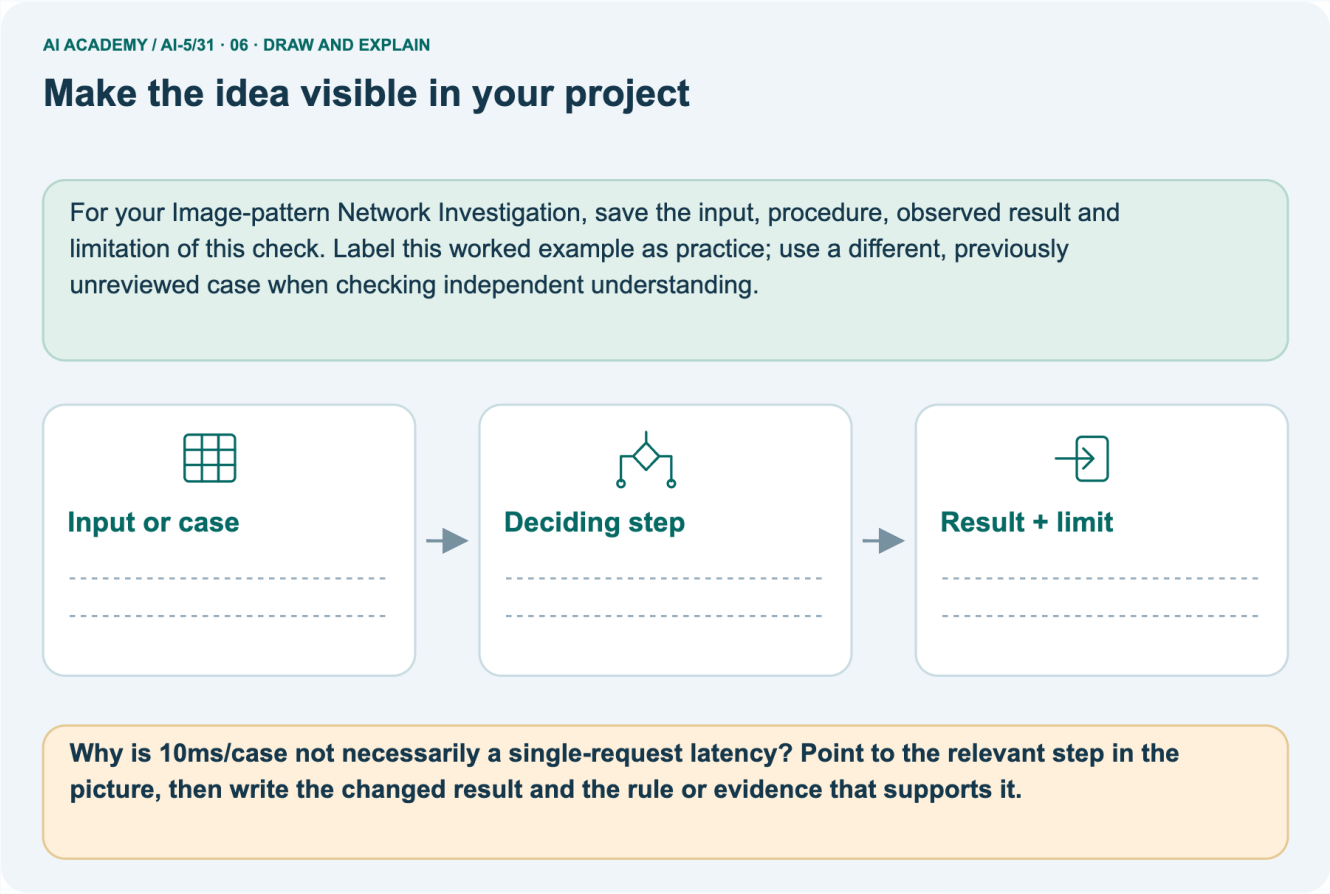
Compare models on the same workload and apply every required condition. P meets both quality and time limits; Q's speed does not compensate for missing the quality gate.

Dividing total run duration by forty gives an amortised time per case. It does not measure the wait for one isolated request, which needs its own timed experiment.

Limit: Record hardware, batching, warm-up and variability; these supplied timings are toy measurements, not actual benchmark runs.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why is 10ms/case not necessarily a single-request **latency**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.